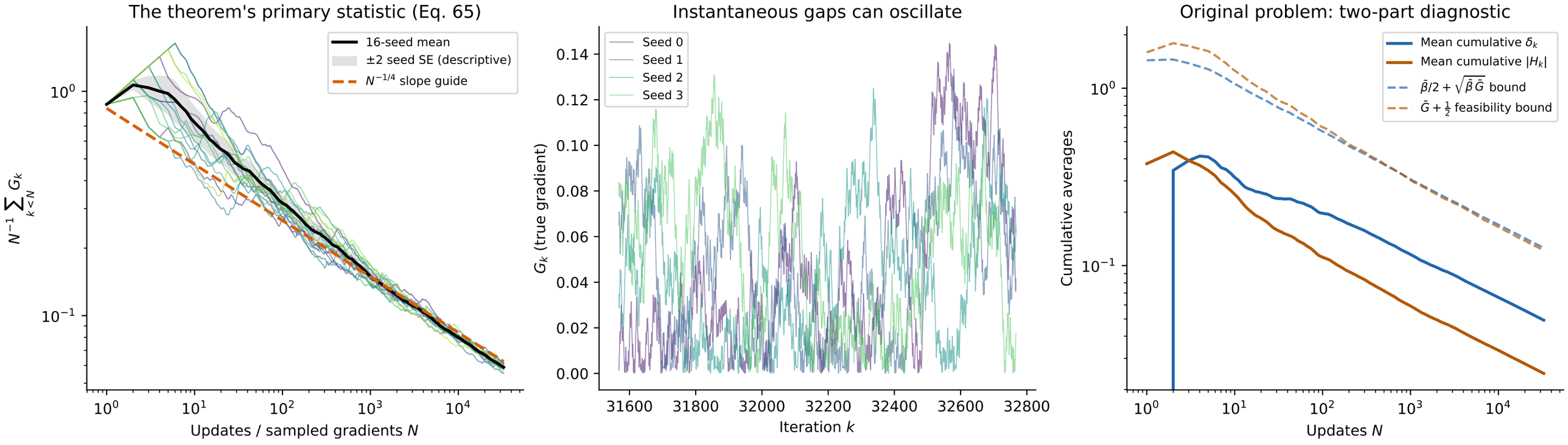

The theorem's primary statistic (Eq. 65)
16-seed mean
±2 seed SE (descriptive)
$N^{-1/4}$ slope guide
$N^{-1}\sum_{k<N} G_k$
Updates / sampled gradients N
Instantaneous gaps can oscillate
Seed 0
Seed 1
Seed 2
Seed 3
G_k (true gradient)
Iteration k
Original problem: two-part diagnostic
Mean cumulative δ_k
Mean cumulative $|H_k|$
$\bar{\beta}/2 + \sqrt{\bar{\beta}\,\bar{G}}$ bound
$\bar{G} + \frac{1}{2}$ feasibility bound
Cumulative averages
Updates N